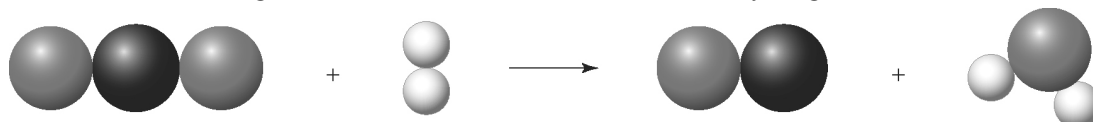


Chemical equations

Answers

Atom and mass conservation

1. Consider the following reaction between carbon dioxide and hydrogen.



Key: Hydrogen (1)

Carbon (12)

Oxygen (16)

- (a) Name the products of this reaction.

Carbon monoxide; water

- (b) Show that the number of atoms of each element is conserved in the reaction.

Reactants: C = 1; O = 2; H = 2

Products: C = 1; O = 2; H = 2

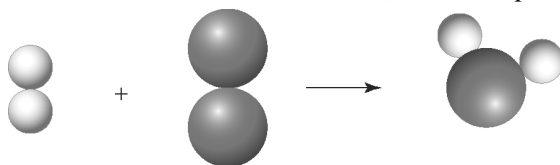
- (c) Use the atomic weight data provided in the key to show that the total mass of reactants equals the total mass of products.

Reactant mass = $(12 + 16 + 16) + (1 + 1) = 46$

Product mass = $(12 + 16) + (1 + 1 + 16) = 46$

Balancing an equation

2. The following diagram models a chemical reaction, but the equation is not balanced.



Key: Hydrogen

Oxygen

- (a) Explain why this equation is not balanced.

The number of atoms of each element in the reactants (H = 2; O = 2) does not equal the number of atoms in the product (H = 2; O = 1).

- (b) Draw in extra molecules in the diagram above to balance the equation being modelled.

